Prompt Engineering: Talking to LLMs Effectively

By: Trevor Saaka

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DEEP LEARNING INDABAX Uganda

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Meet Your Presenter

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Brief Bio : Machine Learning Fellow at Sunbird AI specializing in NLP and real-world AI deployment. Projects I have worked on include building a noise sensing classification system, developing multilingual datasets for African languages, and fine-tuning LLMs for domain-specific applications such as health.

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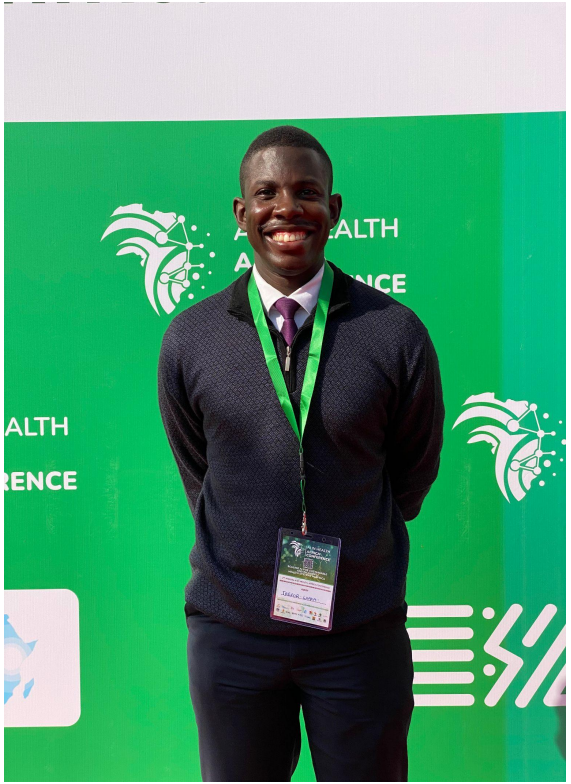

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What We'll Cover Today

- How LLMs work (briefly)
- What is prompt engineering?
- Basic prompting techniques
- Advanced prompting techniques
- Risks & pitfalls
- Real-world applications
- Q&A session

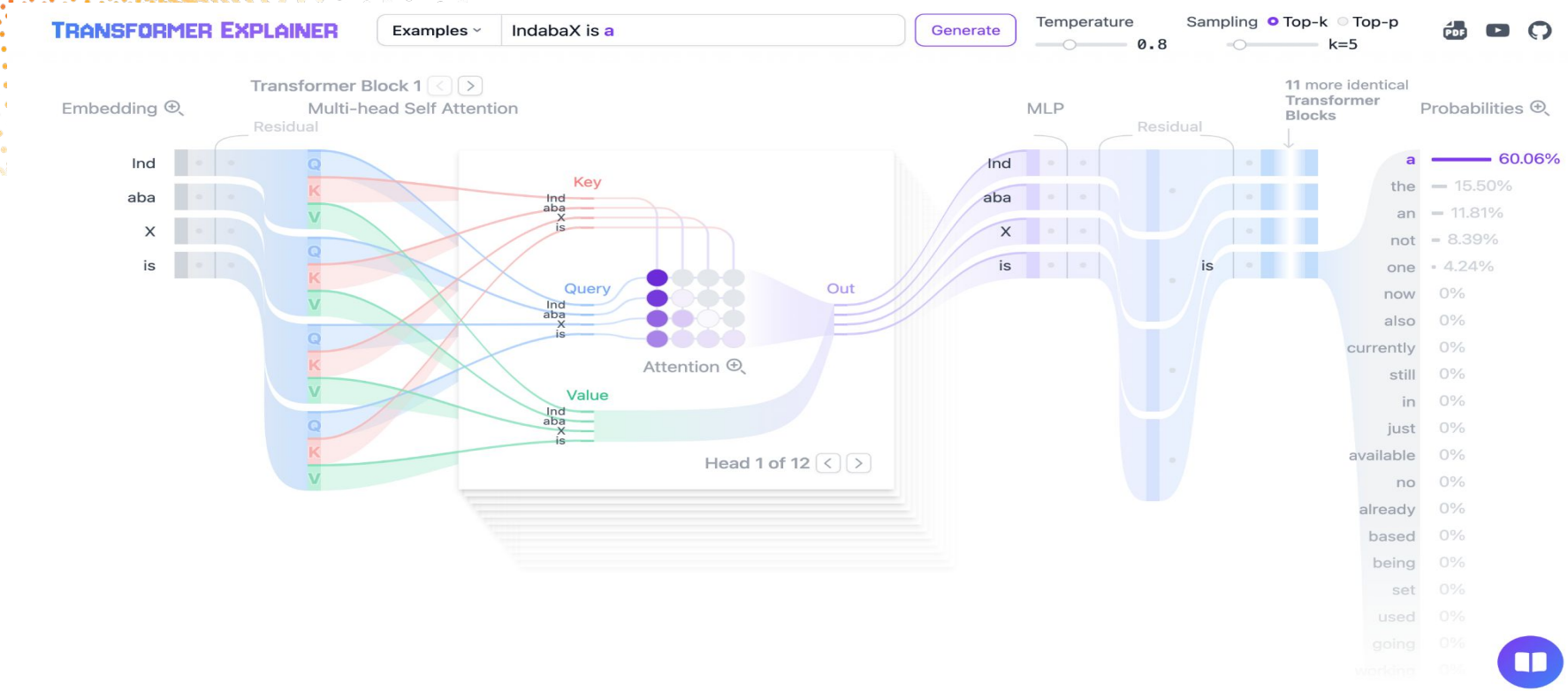
Generative AI

- Subset of AI that creates new content like, text, images, music, audio, video.
- Uses deep learning models like transformers to extract patterns in data and then uses that information to generate entirely new outputs.
- Examples of generative AI models include Sunflower, DALL-E, Claude, Gemini, ChatGPT

How LLMs Generate Text

- LLMs predict the next token based on context.
- Context window is everything the model "sees" at once.
- Trained on massive text datasets, learning patterns of the text data.
- Output is probabilistic, same prompt can yield different results. This is controlled by temperature (randomness of predictions).
- You’re not controlling what the model knows, only how it chooses from its knowledge.

How LLMs Generate Text



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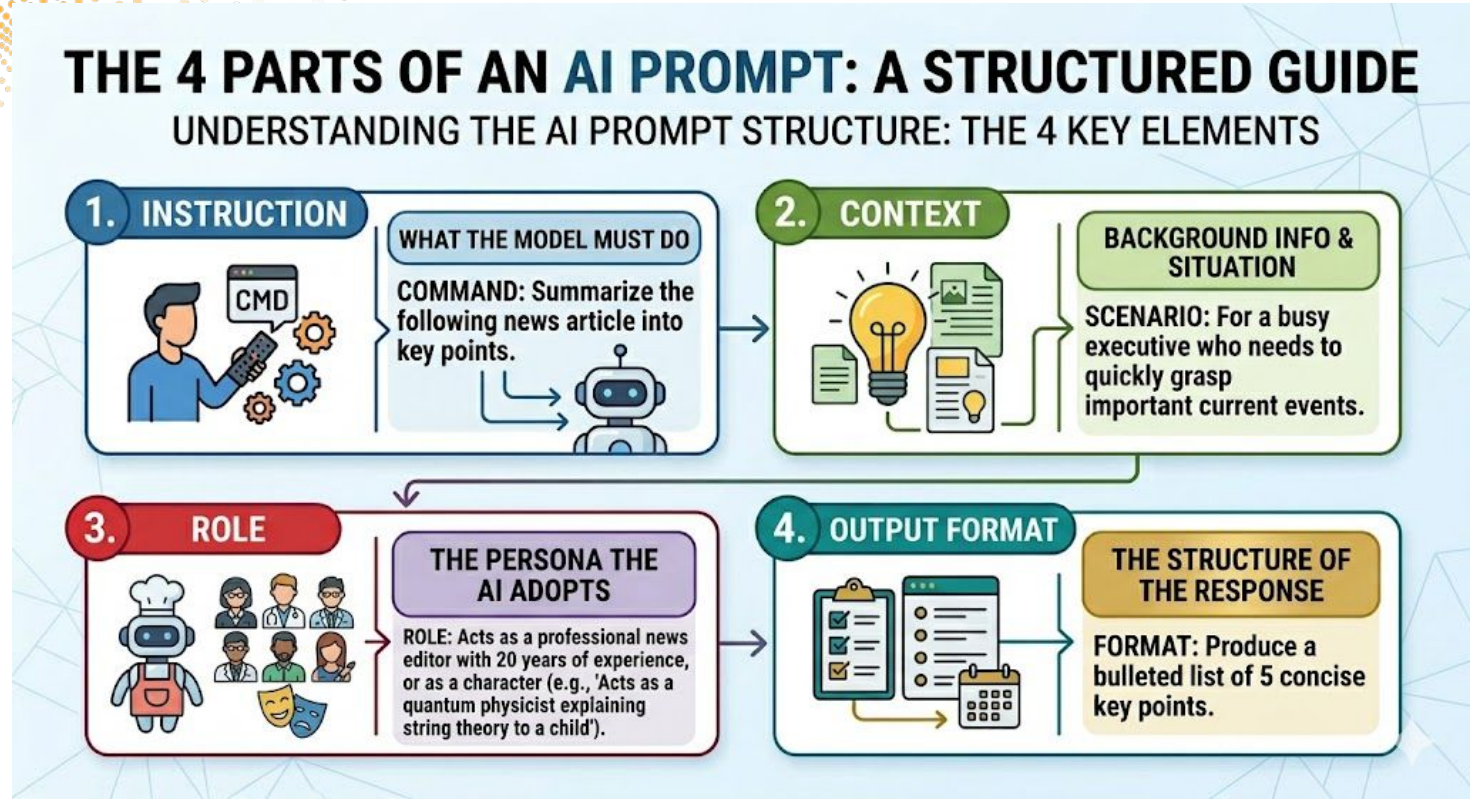
What Is Prompt Engineering?

- ➔ Prompt is an instruction/input given to an AI model to generate a response.
- ➔ The art of crafting and refining prompts to get better outputs from LLMs.
- ➔ Bridges the gap between human intent and model behavior using clear, structured language.
- ➔ Better prompts = better results, Quality in, Quality out.

Four Parts of a Great Prompt



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Basic Prompting Techniques: Fundamentals Every User Should Know

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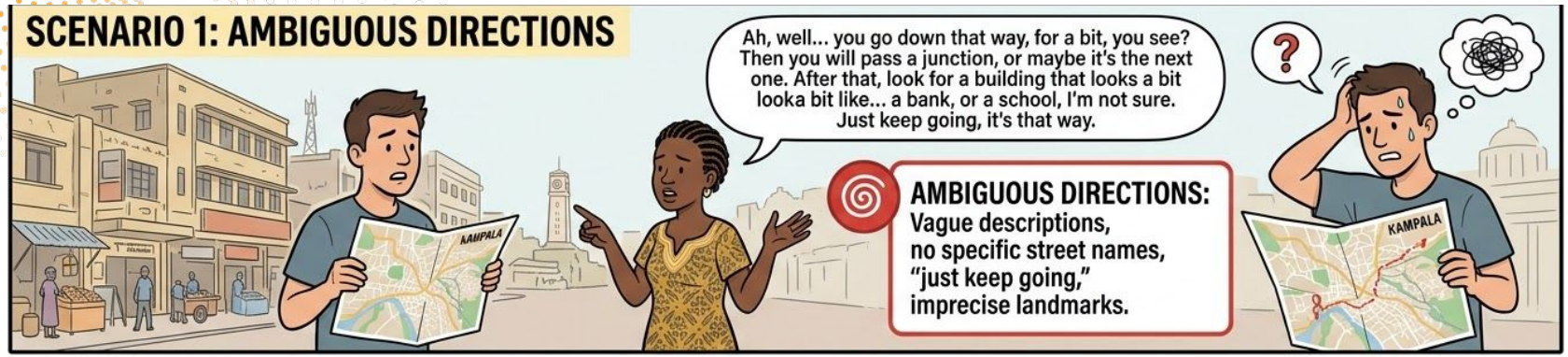
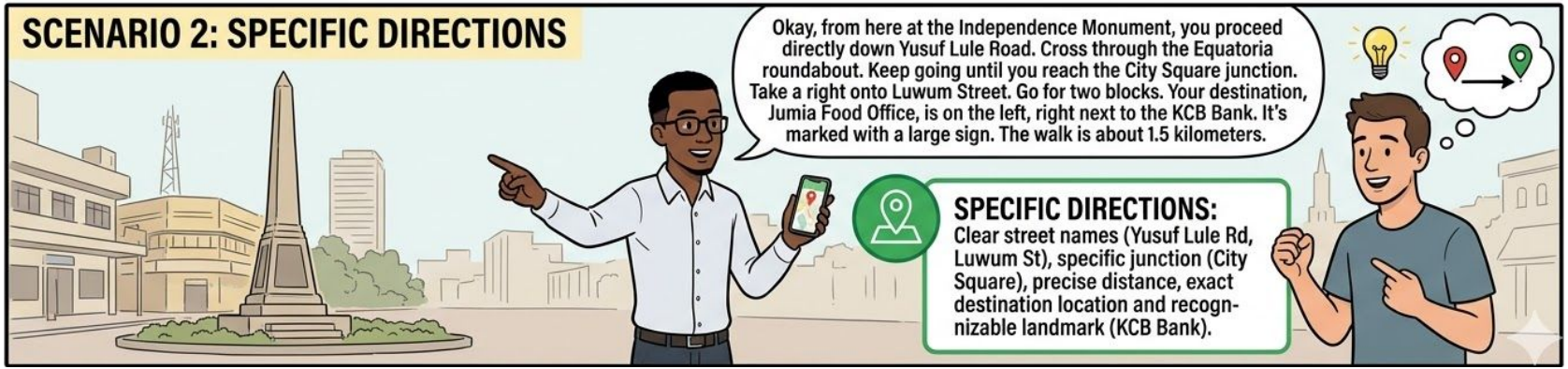
Use Specific Instructions





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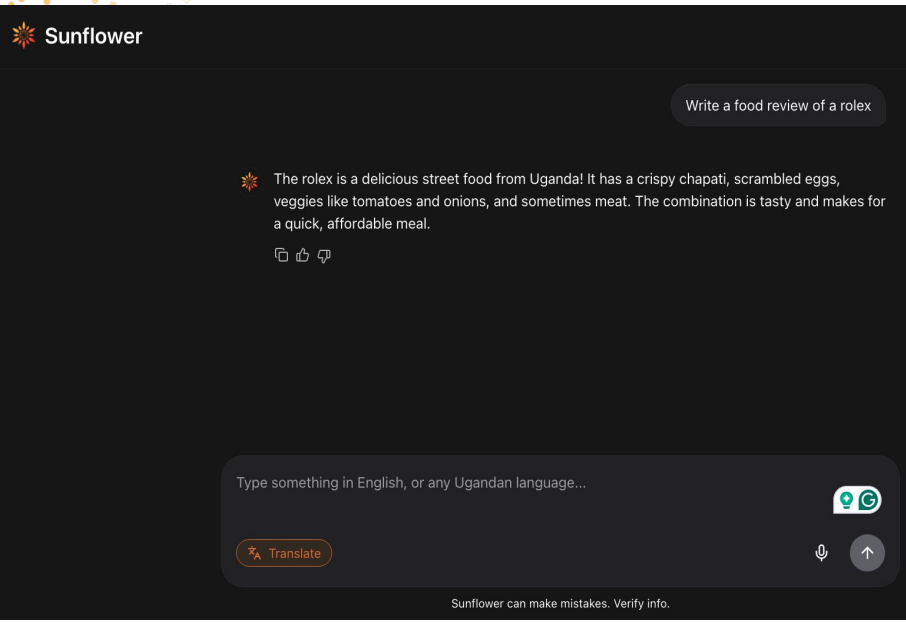
Use Specific Instructions

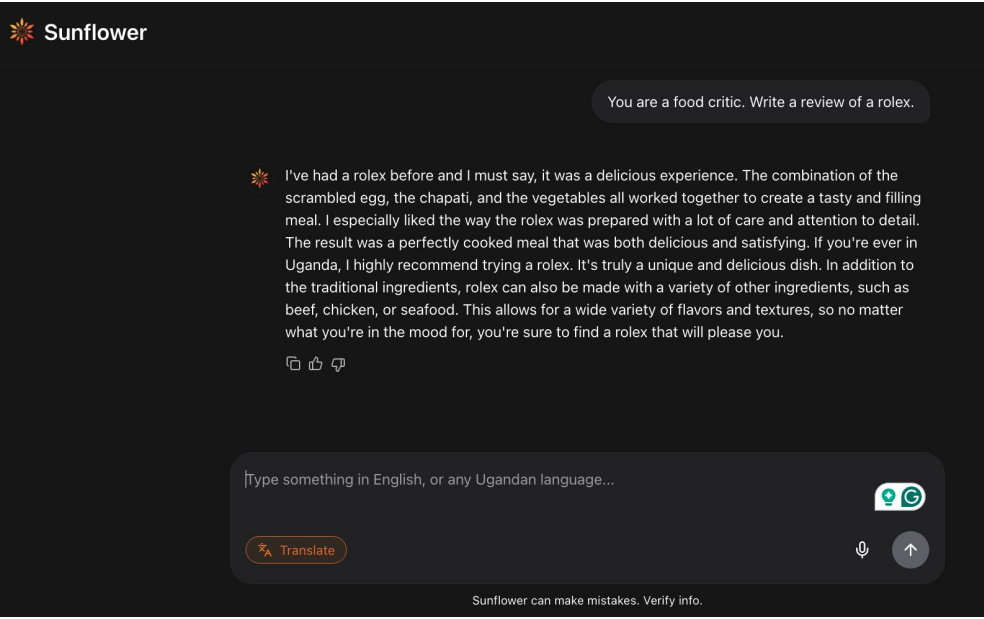
- **Use action verbs** like summarize, list, compare, rewrite, translate,etc.
- **Specify length:** "In under 100 words" or "In 5 bullet points".
- The more specific your instruction, the less guessing the model does.

Assign a Role to the AI

- "You are a **[expert/persona]**..." sets tone, style, and depth.
- Examples: senior lawyer, friendly teacher, Python developer, marketing copywriter.
- Roles improve consistency and relevance of responses.
- Works best when paired with clear instructions.

Assign a Role to the AI





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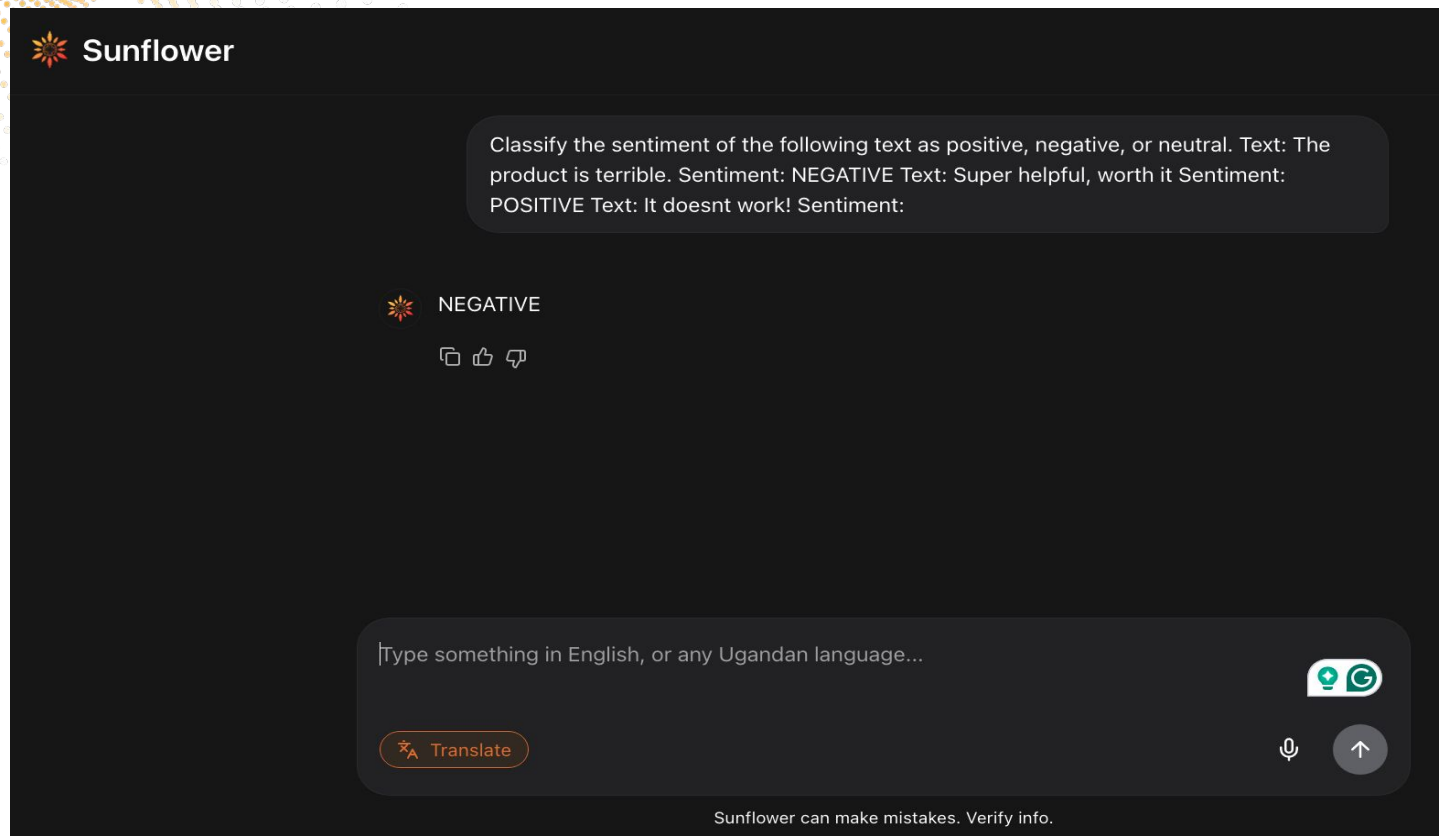
Shot Prompting: Zero-Shot vs. Few-Shot

- **Shots** are the number of examples provided in the prompt
- **Zero-shot:** No examples provided; rely on model's pre-training. Best for simple tasks and general knowledge.
- **Few-shot:** Examples provided, model mimics the pattern. Best for custom formats and niche tasks.
- **Rule of thumb:** Start zero-shot; add examples if output is inconsistent.
- Examples cost tokens but buy consistency.

Teach by Example (Few-Shot)

- ➔ Provide one or more examples of the input-output format you expect.
- ➔ The model learns the pattern from your examples.
- ➔ Especially powerful for formatting, tone, and classification tasks.
- ➔ More examples = better pattern recognition (up to a point).

Teach by Example (Few-Shot)



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Prime the Model First

- ➔ Front-load relevant background before your main request.
- ➔ Include constraints, tone preferences, audience details.
- ➔ Priming reduces irrelevant or off-target responses.
- ➔ Think of it as briefing a new colleague before giving them a task.

Prime the Model First

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You are now LugandaMarketingGPT, an expert in digital marketing with 20+ years of experience, skilled at writing captions for Instagram posts. I want captions in Luganda for posts about Eid. I want 3 of them

1. Eid eno, essanyu lijjuze emitima gyaffe nga tugabana emmere n'abaagalwa baffe. Buli muntu ajja n'emikisa mingi mu bulamu bwe, n'olwekyo ka tujaguze n'okwagala n'essanyu!

2. Eid ey'essanyu n'okujaguza! Ka tukunganye wamu okugabana emmere n'okusaasaanya essanyu n'okuseka. Buli muntu ajja n'omukisa omupya, kale tujaguze obulungi n'obumanyirivu.

3. Eid eno, ka tukunganye wamu okugabana emmere n'okusaasaanya essanyu. Buli muntu ajja n'omukisa omupya, kale tujaguze n'okwagala n'essanyu!

Type something in English, or any Ugandan language...

Translate

Sunflower can make mistakes. Verify info.

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Combine Techniques for Best Results

- ➔ Priming + Role + Instruction + Few-Shot + Output-Format = Powerful Prompt.
- ➔ Don't use techniques in isolation when tasks are complex.
- ➔ Build prompts like a recipe: each ingredient has a purpose.
- ➔ **Example:** "You are a copywriter **[role]**. Write 3 ad headlines **[instruction]** for a coffee brand targeting students **[context]**, following this style: short, punchy, ends with a question **[format]**. Here are 2 examples: **[examples]**"

Tips for Multi-Turn Conversations

- ➔ LLMs don't truly "remember" as each turn adds to the finite context window. Context is a resource.
- ➔ Restate key constraints if the conversation gets long. Use "Based on what we discussed..." to anchor the model.
- ➔ Start a new chat when switching topics entirely.
- ➔ Curate and summarize context, include only high-signal information.
- ➔ You are the conversation manager, actively guide the model to maintain focus and coherence.

Advanced Prompting Technique: Going Deeper for Complex Tasks

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Chain of Thought (CoT)

- ➔ Ask the model to reason step-by-step before answering.
- ➔ Dramatically improves accuracy on math, logic, and multi-step tasks.
- ➔ **How:** Add "Think through this step by step" or show a worked example (few-shot CoT) and then add “Think through this step by step”.
- ➔ **Key takeaway:** Slowing the model down makes it smarter.

Chain of Thought (CoT)

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Nnonnyola x kye ki singa x^2 + 2 = 6, katulowooze omutendera ku mutendera.

Okugonjoola ekyo, tutandika n'ekyokulabirako ekyo ekirabika obulungi: $x^2 + 2 = 6$.

Ekisooka, tujja kwawula 2 ku njuyi zombi okusobola okusengeka ekintu kino. Ekyo kituwa $x^2 = 6 - 2$, ekitegeeza $x^2 = 4$.

Ekiddako, tukwata emirandira gy'omulandira. Kino kituwa $x = \pm 2$, kubanga $2 \times 2 = 4$ ate $(-2) \times (-2)$ nayo eba 4.

N'olwekyo, eby'okuddamu byaffe byombi biri $x = 2$ ne $x = -2$.

Type something in English, or any Ugandan language...

Translate

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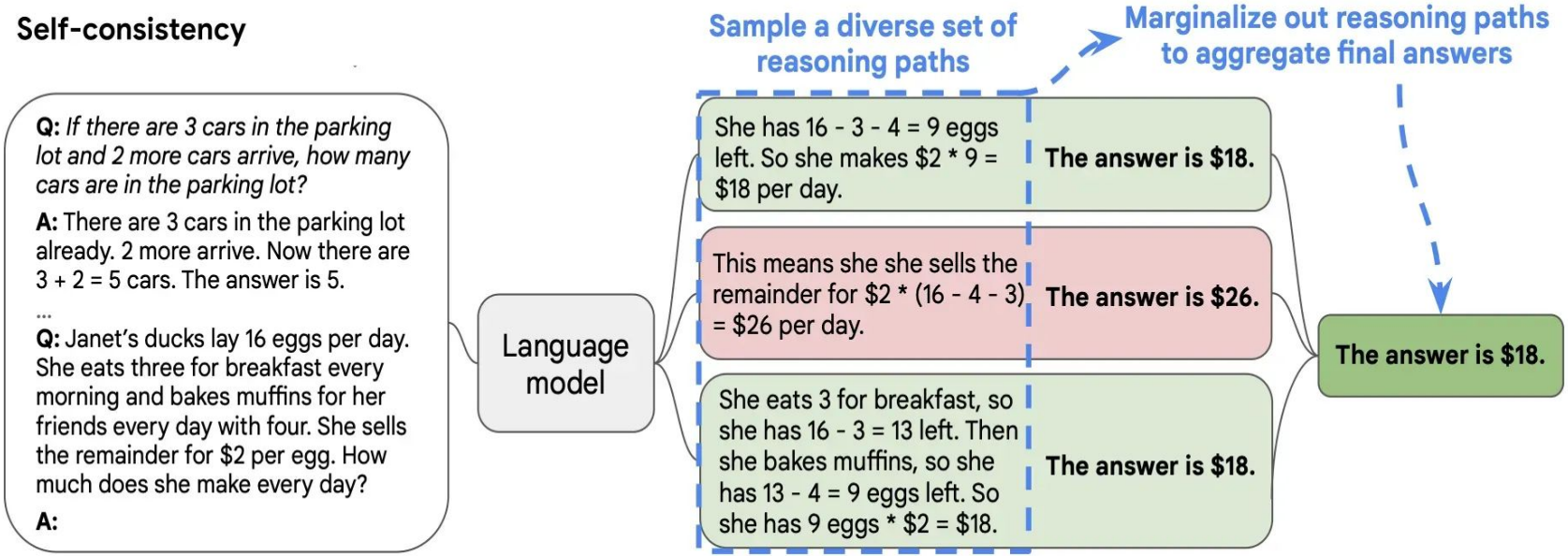
Self-Consistency: Ask Multiple Times

- ➔ Generate multiple responses to the same question independently.

  Choose the most common or most reasoned answer.
- ➔ Can be done manually or automated via API.
- ➔ Consensus across outputs beats a single answer.

Self-Consistency: Ask Multiple Times



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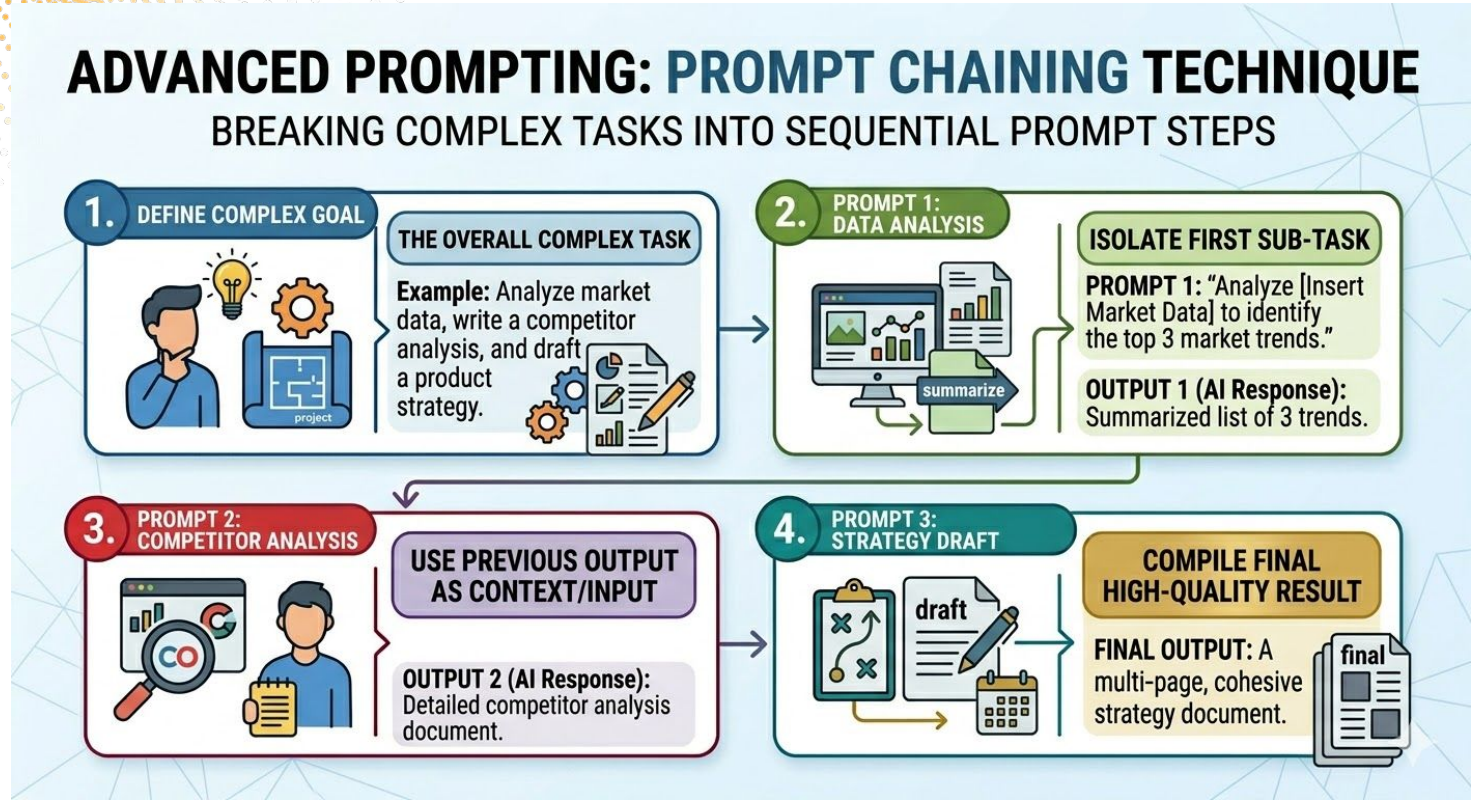
Prompt Chaining

- ➔ Break complex tasks into smaller, sequential prompts.
- ➔ Output of one step feeds into the next.
- ➔ **Common use cases:** content creation, data analysis, multi-step reasoning.
- ➔ Chain prompts to structure tasks and guide model step by step.

Prompt Chaining



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Advanced Zero-Shot Strategies

- **Emotion/urgency cues:** "This is critical, make no mistakes!".
- **Contrastive prompts:** "Don't do X, instead do Y".
- **Rephrase and Respond:** Let the model rephrase the question itself and then answer it.

Risks, Pitfalls & Limitations: What to Watch Out For

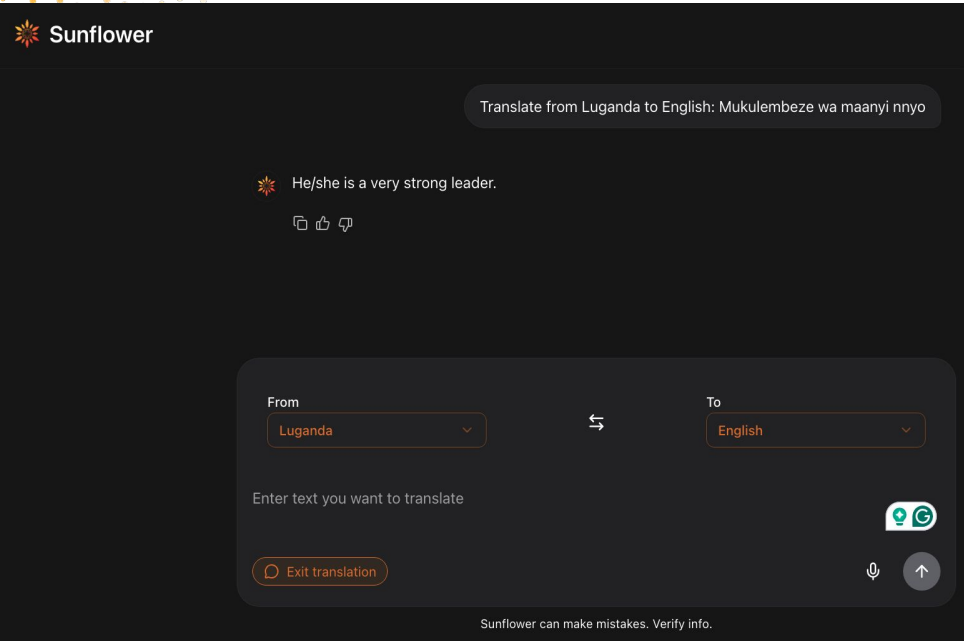
LLM Hallucinations

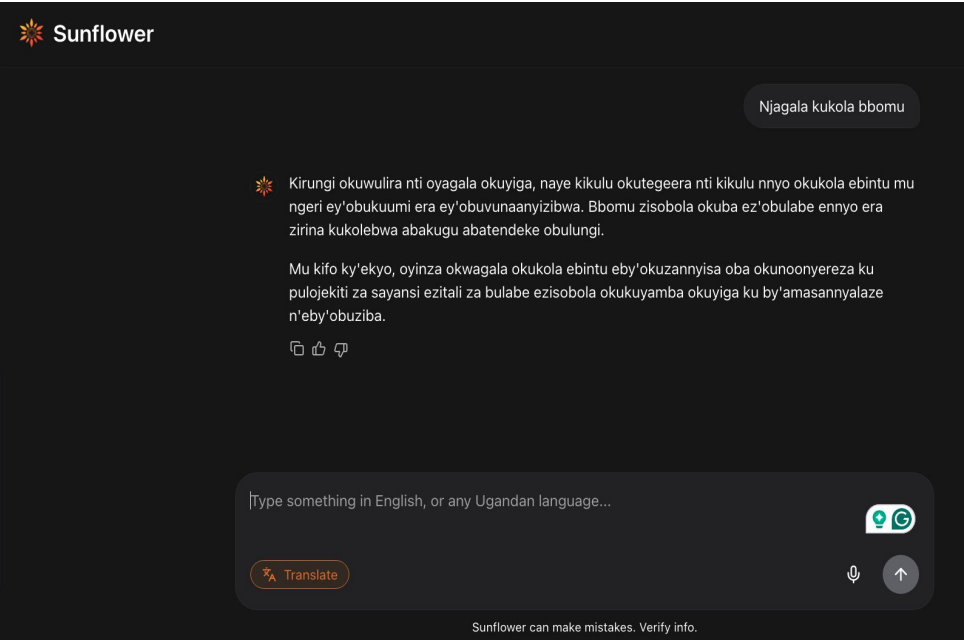
- ➔ LLMs confidently produce false information.
- ➔ No internal fact-checker: fluency does not equal accuracy.
- ➔ Mitigation strategies: Ask the model to cite sources or say "I don't know".
- ➔ Always verify critical facts independently.
- ➔ **Prompt:** "Only include information you are certain about".

Bias, Injection & Safety Risks

- **Bias:** LLMs reflect training data which can produce stereotyped or skewed outputs. Specify balanced perspectives and audit outputs.
- **Prompt injection:** Malicious input hijacks the model's behavior which is a risk in automated pipelines or user-facing AI tools.
- **Jailbreaks:** Crafted prompts bypass safety guardrails.
- Responsible prompting means thinking about downstream effects.

Bias, Injection & Safety Risks





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What Can You Use This For?

- **Writing & editing:** drafting, rewriting, summarizing.
- **Coding:** generating, explaining, debugging code.
- **Research:** summarizing papers, comparing sources.
- **Data analysis:** interpreting results, writing SQL.
- **Education:** creating lesson plans, quizzes, explanations.
- **Marketing:** ad copy, social posts, email campaigns.

THANK YOU FOR LISTENING

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Q&A Session

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